

Supplementary Material: Scalable Bayesian Computational Framework for Decoupling Building Physics from Socioeconomic Constraints at a National Scale

Jules Buckland

S1 Heat Loss Coefficient (HLC) Archetype Baselines

The physical UBEM baseline relies on mapping synthesized households to 32 standardized UK building archetypes. The baseline Heat Loss Coefficient (HLC) values for a subset of the most common archetypes (before floor area adjustment) are presented in Table ??.

Archetype	Built Form	Age Band	Baseline HLC (W/K)
A1	Detached	Pre-1919	420
A2	Detached	1919-1944	380
A3	Semi-Detached	Pre-1919	310
A4	Semi-Detached	1919-1944	270
A5	Mid-Terrace	Pre-1919	230
A6	Mid-Terrace	1919-1944	210
A7	Purpose-Built Flat	Post-1990	90

Table S1: Representative baseline Heat Loss Coefficients (HLC) for standardized UK archetypes.

S2 Iterative Proportional Fitting (IPF) Constraints

The spatial microsimulation relies on Iterative Proportional Fitting (IPF) to generate the synthetic population. The marginal constraints used to align the NEED seed with MSOA-level Census 2021 data are:

- **Tenure:** Owned Outright, Owned with Mortgage, Social Rented, Private Rented.
- **Property Type:** Detached, Semi-Detached, Terraced, Flat/Maisonette.
- **Age Band:** Pre-1919, 1919-1944, 1945-1980, Post-1980.
- **Household Income Decile:** Derived from the local Index of Multiple Deprivation (IMD) Income Domain score.

S3 BYM2 Prior Specifications

The Bayesian spatial model utilizes the BYM2 (Besag-York-Mollie 2) specification. The full prior structure is detailed below:

$$\begin{aligned}
y_{m,h} &\sim \text{Normal}(\mu_{m,h}, \sigma_y^2) \\
\mu_{m,h} &= \log(T_{m,h}) + \beta_{th} + \beta_{inc}Z_{inc,m} + \sigma_{total}(\sqrt{1-\rho}\theta_m + \sqrt{\rho}\phi_m) \\
\beta_{th} &\sim \text{Normal}(-0.3, 0.1) \\
\beta_{inc} &\sim \text{Normal}(0, 0.5) \\
\sigma_y &\sim \text{HalfNormal}(0.5) \\
\sigma_{total} &\sim \text{HalfNormal}(1.0) \\
\rho &\sim \text{Beta}(0.5, 0.5) \\
\theta_m &\sim \text{Normal}(0, 1) \\
\phi_m &\sim \text{ICAR}(\mathbf{W})
\end{aligned}$$